

Institute of Actuaries of India

Subject ST6 – Finance and Investment B

May 2012 Examination

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1 :

- Let p_1 and c_1 be prices of put and call option with strike price K_1 and similarly p_2/c_2 for strike price K_2 . Let $K_2 > K_1$. From put-call parity:
 - $p_1 + S_0 = c_1 + K_1 * e^{-rt}$
 - $p_2 + S_0 = c_2 + K_2 * e^{-rt}$
 - $p_1 - p_2 = c_1 - c_2 - (K_2 - K_1) * e^{-rt}$
- Therefore, the cost of setting up the bull spread using put option would be lower than the cost incurred using call option by $(K_2 - K_1) * e^{-rt}$. The profit when calls are used to create the spread is higher than when put is used by $(K_2 - K_1) * (1 - e^{-rt})$. This reflects the fact that using the call involves additional risk free investment of $(K_2 - K_1) * e^{-rt}$ over the put strategy.

[5]**Solution 2 :****(a)**

- The number of short futures contracts required = $\frac{500,000,000 \times 5.0}{1.20 \times 500,000 \times 8} = 520.83$
- 581 contracts should be shorted.

(b)

- The number of short futures contracts required = $\frac{500,000,000 \times 5.0}{1.20 \times 500,000 \times 6} = 694.44$
- 694 contracts should be shorted.

(c)

- In this case the gain on the short futures position is likely to be less than the loss on the bond portfolio. This is because the gain on the short futures position depends on the size of the movement in long-term rates and the loss on the bond portfolio depends on the size of the movement in short-term rates and medium-term rates. Duration-based hedging assumes that the movements in the two rates are the same.

[6]

Solution 3 :

- The price of Indian rupee expressed in terms of US dollar is $\frac{1}{P_{\$}}$

$$G = \frac{1}{P_{\$}}$$

- Assume

$$\frac{\partial G}{\partial t} = 0, \quad \frac{\partial G}{\partial P_{\$}} = -\frac{1}{P_{\$}^2}, \quad \text{and} \quad \frac{\partial^2 G}{\partial S^2} = \frac{2}{P_{\3$

- From Ito lemma, we have

$$dG = \left[-\frac{1}{P_{\$}^2} (r_I - r_{US}) P_{\$} + \frac{1}{2} \frac{2}{P_{\$}^3} \sigma^2 P_{\$}^2 \right] dt - \frac{1}{P_{\$}^2} \sigma P_{\$} dz$$

$$dG = \frac{1}{P_{\$}} [-(r_I - r_{US}) + \sigma^2] dt - \frac{1}{P_{\$}} \sigma dz$$

$$dG = [-(r_I - r_{US}) + \sigma^2] G dt - \sigma G dz$$

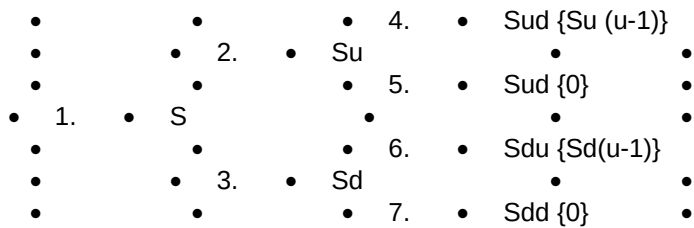
$$dG = [r_{US} - r_I + \sigma^2] G dt + \sigma G dz^*$$

- Where $dz^* = -dz$

- We would have expected the growth rate for $1/P_{\$}$ to be $r_{US} - r_I$ but we find an additional term involving the volatility. It may be noted that the numeraire for equation given in question is money market account in INR and since the equation for $1/P$ is derived using the same equation the numeraire remains the same. To be symmetrical the numeraire for $1/P_{\$}$ needs to be changed from money market account in INR to money market account in USD. If we change the numeraire to USD then we would find that the equation is symmetric.

[6]**Solution 4 :**

- Please note that under standard binomial model the length of the two periods are equal i.e. $T_0 = T_1/2$. Let B denote the discounted value of one rupee over one time period i.e. $B = e^{-rT_0} = e^{-r(T_1 - T_0)}$. Under this model the risk-neutral probability of stock price moving up remains same for both the periods and it is $q = \frac{B^{-1} - d}{u - d}$. The discounted stock price is martingale under the above measure under standard binomial model. The conditions given in the question are required for the formula of the risk neutral probability to lie within 0 and 1.
- The stock price tree along with the option payoff at maturity of the underlying option is given below.



- It may be noted that though node 5 and 6 are same for the stock price but they are not same for option pay-offs. The value of the option at node 2 is given below:
- $BqSu(u-1)$ and node 3 is $BqSd(u-1)$. Therefore the value of the option at node 1 would be the discounted risk neutral expected value as given below:
- $B\{q \cdot BqSu(u-1) + (1-q) \cdot BqSd(u-1)\}$
- $=B^2qS(u-1)\{qu+d(1-q)\}$
- $=B^2qS(u-1)\{q(u-d)+d\}$
- $=B^2qS(u-1)\{B^{-1}\}$ [substituting the value of q]
- $=BqS(u-1)$
- $=B \cdot (q \cdot (Su-S) + (1-q) \cdot 0)$
- We found that the value of the option is same as at the money option today for one period (T1-T0). The crucial factor in this result would be the assumption that interest rates are flat and known in advance.

[7]

Solution 5 :**(a)**

Year	Interest rate	discount factor
1	8.00%	0.9259
2	7.50%	0.8653
3	7.00%	0.8163
4	6.50%	0.7773
5	6.00%	0.7473
sum		4.1321
swap rate		6.116%

(b)

- Company can match the duration of payout by combination of short term bonds and forward swaps. In this case company can buy a 20 year bond and a forward swap starting at year 20 till year 30 .

(c)

- If the spread between the gilt and swap increases, then value of liability will remain the same but market value of receiver swaps will go down due to increase in swap rate. This will lead to over all fall in net assets (assets-liability)

(d)

Variation of the balance sheet due to the swap spread can be hedge entering a contract to purchase a gilt one year from now the price would be the one year forward price of the gilt; and

Simultaneously entering a swap starting in one year – with the same maturity as the gilt – under which the insurer pays fixed and receives floating.

This trade would be cash settled one year from now. If swap spreads widen over the next year then this hedging trade will deliver a profit which can be used to (partly) offset the loss experienced on the insurer's balance sheet.

•	• increase in swap spread	• decrease in swap spread
• Forward Gilt	• No Change	• No Change
• Forward Payer swap	• Increase in value	• Decrease in value
• Liability	• No Change	• No change
• Receiver swap	• Decrease in value	• Increase in value

[11]**Solution 6 :****(a)**

- A synthetic CDO is a form of collateralized debt obligation (CDO) in which the underlying credit exposures are taken on using a credit default swap rather than by having a vehicle buy assets such as bonds.
- Synthetic CDOs package up the default losses on a reference portfolio of credit names into a number of tranches. Each tranche is specified by its “attachment points” – the range over which the tranche is responsible for losses on the reference portfolio

(b)

- If there are j defaults in the portfolio by time T_i , the total loss due to default is $A^*(1-\delta)$. The tranche with attachment points *Upper* and *Lower* is only responsible for losses exceeding $NA \times \text{Lower}$ but no greater than $NA \times \text{Upper}$ i.e. this tranche is responsible for losses amounting to:

- $Tranche\ loss(Lower, Upper) = NA \times \max\left(\min\left(\frac{j}{N} \times (1 - \delta), Upper\right) - lower, 0\right)$
- Risk neutral expectation of the loss at time period T is
- $EL(T_i) = NA \times \sum_{j=0}^N P_{defaults}(j, T_i) \times \max\left(\min\left(\frac{j}{N} \times (1 - \delta), Upper\right) - lower, 0\right)$
- Losses are incurred incrementally i.e. at time T_i losses up until time T_{i-1} have already been accounted for and the contingent cash flow at time T_i is just the incremental loss over the period from T_{i-1} to T_i . Incremental expected loss at time T_i is $EL(T_i) - EL(T_{i-1})$ and hence the total value of the contingent leg is
- $Contingent\ leg = \sum_{i=1}^n DF(T_i) \times (EL(T_i) - EL(T_{i-1}))$
- where $DF(t)$ represent discount factor for time t .
- The fee leg also takes into account the reduction in principal as default losses affect the tranche. The tranche principal at initiation is $NA \cdot (Upper - Lower)$ and this is reduced by default losses as they occur i.e. the expected tranche principle outstanding on payment date T_i is $NA(Upper - Lower - EL(T_i))$. Hence, if the spread payment is $Spread$ per annum of outstanding principal the total value of the fee leg is:
- $Fee\ leg = Spread \times \sum_{i=1}^n (T_i - T_{i-1}) DF(T_i) \times (NA \times (Upper - Lower) - EL(T_i))$
- As at the start of CDO it is priced as Zero, hence all the cash flow should balance out. Thus
- $Spread = \frac{\sum_{i=1}^n DF(T_i) \times (EL(T_i) - EL(T_{i-1}))}{\sum_{i=1}^n (T_i - T_{i-1}) DF(T_i) \times (NA \times (Upper - Lower) - EL(T_i))}$

(c)

- The key input into the calculation of the spread for a given CDO tranche is thus the risk-neutral distribution of the number of losses on the underlying reference portfolio, from which we can calculate the expected loss at each payment date. The number of losses is in turn determined by the joint risk-neutral probability of default on all names in the portfolio
- The standard model is used by market to determining this joint probability distribution of the Spread and this model is called "Gaussian Copula"

[11]

Solution 7 :

- The price of the bond at time t is $P = e^{-Y(T-t)}$
- $\frac{\partial P}{\partial Y} = -(T - t)e^{-Y(T-t)}$

- From Ito lemma, the volatility of the bond price is:
- $\sigma \frac{\partial P}{\partial Y} = -\sigma(T-t)e^{-Y(T-t)}$ which tends to zero as $t \rightarrow T$.

[3]**Solution 8 :****(a)**

- The sum on the left hand side is a telescoping sum which reduces to
- $\sum_{j=1}^n [t_j W_{t_j} - t_{j-1} W_{t_{j-1}}] = t_1 W_{t_1} - t_0 W_{t_0} + t_2 W_{t_2} - t_1 W_{t_1} \dots + t_n W_{t_n} - t_{n-1} W_{t_{n-1}} = t_n W_{t_n} - t_0 W_{t_0} = t W_t$
- since $t_n = t$ and $W_{t_0} = W_0 = 0$.
- The right hand side is also a telescoping sum which can be simplified to:
- $\sum_{j=1}^n [t_j (W_{t_j} - W_{t_{j-1}})] + \sum_{j=1}^n [(t_j - t_{j-1}) W_{t_{j-1}}] = t_1 W_{t_1} - t_1 W_{t_0} + t_1 W_{t_0} - t_0 W_{t_0} + t_2 W_{t_2} - t_2 W_{t_1} + t_2 W_{t_1} - t_1 W_{t_1} + \dots + t_n W_{t_n} - t_n W_{t_{n-1}} + t_n W_{t_{n-1}} - t_{n-1} W_{t_{n-1}}$
- $= t_n W_{t_n} - t_0 W_{t_0} = t W_t$
- In this example with only one process being stochastic, the above does represent the standard product rule. Let $u(t) = t$ and $v(t) = W_t$. An extra term would account for the quadratic variation if the integrand and the integrator were both semimartingales.

(b)

- In the limit the partition width goes to zero and the above converges to integral and we can deduce that:
 - $\text{LHS} = \sum_{j=1}^n [t_j W_{t_j} - t_{j-1} W_{t_{j-1}}] \rightarrow \int_0^t d(s W_s)$
- $$t W_t = \int_0^t s dW_s + \int_0^t W_s ds \Rightarrow \int_0^t s dW_s = t W_t - \int_0^t W_s ds$$

(c)

- The integral $\int_0^t s dW_s$ is defined in the sense of Ito .

[8]

Solution 9 :

(a)

- A *chooser* option (sometimes referred to as an *as you like it* option) has the feature that, before a specified period of time (T_1), the holder can choose whether the option is a call or a put. The underlying call/put option would have the maturity period T_2 which would be longer than T_1 .
- The value of the chooser option at time t (the exercise time) would be maximum of the call or put value since it can be converted into any one of these at this point in time.

(b)

- $C(S_t, t) - P(S_t, t) = e^{-r(T-t)} E[(\max(S_T - K, 0) - \max(K - S_T, 0)) | \mathcal{F}_t]$
- $= e^{-r(T-t)} E[(\max(S_T - K, 0) + \min(S_T - K, 0)) | \mathcal{F}_t] = e^{-r(T-t)} E[(S_T - K) | \mathcal{F}_t]$
- $= S_t - e^{-r(T-t)} K$ since under the risk neutral measure, the discounted stock is a martingale. This relationship is put-call parity.

- Using the solution derived in (a) above, it follows that at time t ,

$$H(S_t, t) = \max(C(S_t, t), P(S_t, t))$$

Substituting for Put value from the above equation would give the desired equation.

$$H(S_t, t) = \max(C(S_t, t), C(S_t, t) + e^{-r(T-t)} K - S_t)$$

(c)

- From $H(S_t, t) = \max(C(S_t, t), C(S_t, t) + e^{-r(T-t)} K - S_t)$ above,
- $H(S_t, t) = \max(C(S_t, t), C(S_t, t) + e^{-r(T-t)} K - S_t) = C(S_t, t) + \max(0, e^{-r(T-t)} K - S_t)$
- Therefore, $H(S_0, 0) = e^{-rt} E[C(S_t, t) + \max(0, e^{-r(T-t)} K - S_t)]$
- $= e^{-rt} C(S_0, 0) e^{rt} + e^{-rt} E[\max(0, e^{-r(T-t)} K - S_t)]$
- $= C(S_0, 0) + e^{-rt} E[\max(0, K - e^{r(T-t)} S_t)]$
- $= C(S_0, 0) + e^{-rT} E[\max(0, K - e^{r(T-t)} S_t)]$
- Note that $S_t = S_0 e^{(r - \frac{1}{2}\sigma^2)t + \sigma W_t}$. Substituting this in the above we get the equation required.

(d)

- Since $e^{-rT} E[\max(0, K - e^{r(T-t)} S_t)] = e^{-r(T-t)} E[\max(0, e^{-r(T-t)} K - S_t)]$ the chooser option is equivalent to a portfolio consisting of a long call expiring at T with strike K plus a long put with strike $e^{-r(T-t)} K$ expiring at t . Therefore,
- $H(S_0, 0) = C(S_0, 0, K, T) + P(S_0, 0, e^{-r(T-t)} K, t)$

(e)

- Therefore, using the above,
- $H(S_0,0)=S_0 \left(N(d_1) - N(-\bar{d}_1) \right) + K e^{-r(t-T)} \left(N(-\bar{d}_2) - N(d_2) \right)$
- Where $d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$
- $d_2 = d_1 - \sigma\sqrt{T}$
- $\bar{d}_1 = \frac{\ln\left(\frac{S_0}{K}\right) + rT + \frac{1}{2}\sigma^2 t}{\sigma\sqrt{t}}$
- $\bar{d}_2 = \bar{d}_1 - \sigma\sqrt{T}$

[15]**Solution 10 :**

- (a)** The numeraire for a probability measure is a function (entity) with the property that any tradable asset divided by it is a martingale under that measure. Risk neutral measure have cash roll up as numeraire , thus price of any tradable asset divided by cash roll up should be martingale.

(b)

- Forward rate at time between the time period S and T is defined in terms of $P(t,T/S)$
- $F(t;T,S) = \frac{1}{S-T} \left(\frac{P(t,T)}{P(t,S)} - 1 \right)$

(c)

- We can associate a measure with each forward rate. We define the forward measure for the maturity $T_k + \delta$ as the measure which has the zero coupon bond maturing at $T_k + \delta$ as its numeraire, and denote this measure by Q_k . This simply means that the price of any tradable asset, divided by $P(t, T_k + \delta)$, is a martingale under Q_k . In particular, the portfolio of zero coupon bonds:
- $\left(\frac{1}{\delta}\right) \times (P(t, T_k) - P(t, T_k + \delta))$
- is a tradable asset. Dividing through by $P(t, T_k + \delta)$ gives the forward rate $F_k(t)$, which is therefore a martingale under Q_k . Forward rates are martingales under their associated forward measure.

Hence under this martingale measure we can express the forward rate by differential equation with zero drift

- $dF_k(t) = F_k(t) \sum_{q=1}^n \sigma_k^q dZ^q$
- where $Z_q(t)$ are independent Brownian motions and σ_k^q is the component of volatility of $F_k(t)$ due to the q 'th factor Z_q . n is the number of factors
- The forward rates are drift less only under their particular forward measures

(which are different for each forward rate). Under the risk neutral measure forward rate can be expressed as

$$\frac{DF_k(t)}{F_k(t)} = \sum_{j=m(t)}^k \frac{\delta F_j(t) \sum_{q=1}^n \xi_j^q(t) \xi_k^q(t)}{1 + \delta F_j(t)} dt + \sum_{q=1}^n \xi_k^q(t) dZ^q(t)$$

(d)

- This formulation of forward rates in risk neutral measure is representation of the Libor Market model
- Advantage
- Positive interest rate
- Flexibility in specifying volatility structure
- Models the forward rate which is a tradable quantity
- Disadvantages
- Lognormal projection of forward rate leads to exploding scenarios
- No close form solution of option prices

[12]

Solution 11 :

- The delta of the option is the rate of change of the value of the option with respect to the price of the asset. When the asset price increases by a small amount the value of the option decrease by 30 times this amount. The gamma of the option is the rate of change of their delta with respect to the price of the asset. When the asset price increase by a small amount the delta of the portfolio decrease by 5 times this amount.
- Suppose δS is the change in the asset price in one day and δx is the proportional change in the asset price. The quadratic model is
- $\delta P = -30 \delta S - 0.5 * 5 * (\delta S)^2 = -30 * 20 \delta x - 0.5 * 5 * 20^2 * (\delta x)^2$ which simplifies to $\delta P = -600 \delta x - 1000 * (\delta x)^2$
- The first three moments of the portfolio value are
- $0.5 * 20^2 * -5 * 0.01^2 = -0.1$
- $20^2 * 30^2 * 0.01^2 + \frac{3}{4} * 20^4 * 5^2 * 0.01^4 = 36.03$
- $\frac{9}{2} * 20^4 * 30^2 * -5 * 0.01^4 + \frac{15}{8} * 20^6 * (-5)^3 * 0.01^6 = -32.415$
- Therefore, the mean change in the portfolio value in one day is -0.1 and the standard deviation of the change in one day is $\sqrt{36.03 - 0.1^2} = 6.002$
- The skewness is
- $\frac{-32.415 - 3 * 36.03 * (-0.1) + 2 * (-0.1)^3}{6.002^3} = \frac{-21.608}{6.002^3} = -0.10$

- Using only the first two moments the 1-day 99% VAR is therefore $0.1 + 2.33 * 6.002 = 14.08$
- When the three moments are considered we can use the Cornish Fisher expansion to estimate the VAR which is given by
- $-0.1 - 6.002w_q$ where
- $w_q = -2.33 + \frac{1}{6}(2.33^2 - 1) * (-0.1) = 2.404$
- Therefore, the one percentile point is $-0.1 - 6.002 * 2.404 = -14.529$ giving us a slight increase from our earlier estimate.

[6]

Solution 12 :**(a)**

- We can create a synthetic forward bid contract by doing transactions in money market and spot market. So, if we want to sell one dollar in future (1 year later) than we borrow present value of 1 USD today, convert it into rupees using the spot market and then deposit the rupees for one year. This would mean that we would need to repay one dollar in future and we would get a certain amount of rupees from our investment. This would give us the synthetic forward rate as
- $\frac{1}{1+i_{\$a}} * S_b * (1 + i_b) = F_{sb} = \frac{1}{1+0.031} * 50 * (1 + 0.09) = 52.8613$
- Where $i_{\$}$ stands for USD interest rate, i for Rupee interest rate, S the spot rate. The subscript b or a stands for bid/ask rate and subscript s stands for synthetic rate. Similarly $F_{sa} = \frac{1}{1+i_{\$b}} * S_a * (1 + i_a) = \frac{1}{1+0.03} * 50.1 * (1 + 0.0925) = 53.1401$. Therefore the synthetic buy/sell rate is (52.86, 53.14). Since, the actual forward buy rate is 52.25 which is lower than the synthetic sell rate 52.8613, one can carry out the arbitrage to create a synthetic sell portfolio and buy the forward contract. This would give rise to a net profit of 0.5113 rupee per dollar. This is riskless profit and hence a pure arbitrage opportunity.
- Since a lot of buying of the forward contract would take place and sell of usd in the spot market would also take place would imply that forward rates would go up and spot rates for dollar would go down. There can also be impacts on the interest rates in these markets. However, the rates would re-align so that such opportunity does not exist.

(b)

- One can invest USD directly in the US money market or one may invest the same in the Indian money market. The investment return on the US money market would be 3%. To invest in the Indian money market, one would have to convert it into rupees, invest the rupees and buy a forward contract to protect against the exchange rate movement. The effective return would be $50 * \frac{1.09}{52.87} = 1.0308$ i.e. effective interest of 3.08%. So, by investing in INR money market the yield would go up by 8bps.

(c)

- It may be noted that since the revised forward rate interval overlaps with the

synthetic forward rate pure arbitrage opportunity does not exist. However, one can still increase the yield on investment or borrow cheaper which is called one way arbitrage.

- Arbitrage opportunities may persist if the market forces are not allowed to operate due to various reasons like control on short selling, control on participants in the market, limit on borrowing/lending in foreign currency etc.
- Arbitrage may also persist due to other market protection related market mechanisms like margin, allowance for credit risk, transaction costs, taxation etc. The taxation regime may create differences on the transactions related to foreign currency compared to Indian currency.

[10]

[Total Marks – 100]
